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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 9

Title: Exploratory data analysis on business datasets

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Exploratory Data Analysis on Business Datasets

Problem Statement

You are a Data Analyst at a consulting firm. A client has provided a large business dataset and wants to understand key trends, patterns, and relationships within the data. The objective is to perform Exploratory Data Analysis (EDA) to examine the dataset, identify meaningful business insights, detect patterns and outliers, analyze key performance indicators, and discover opportunities for improving business performance and supporting data-driven decision-making.

Dataset Used

Superstore Sales Dataset / Business Sales Dataset (Kaggle)

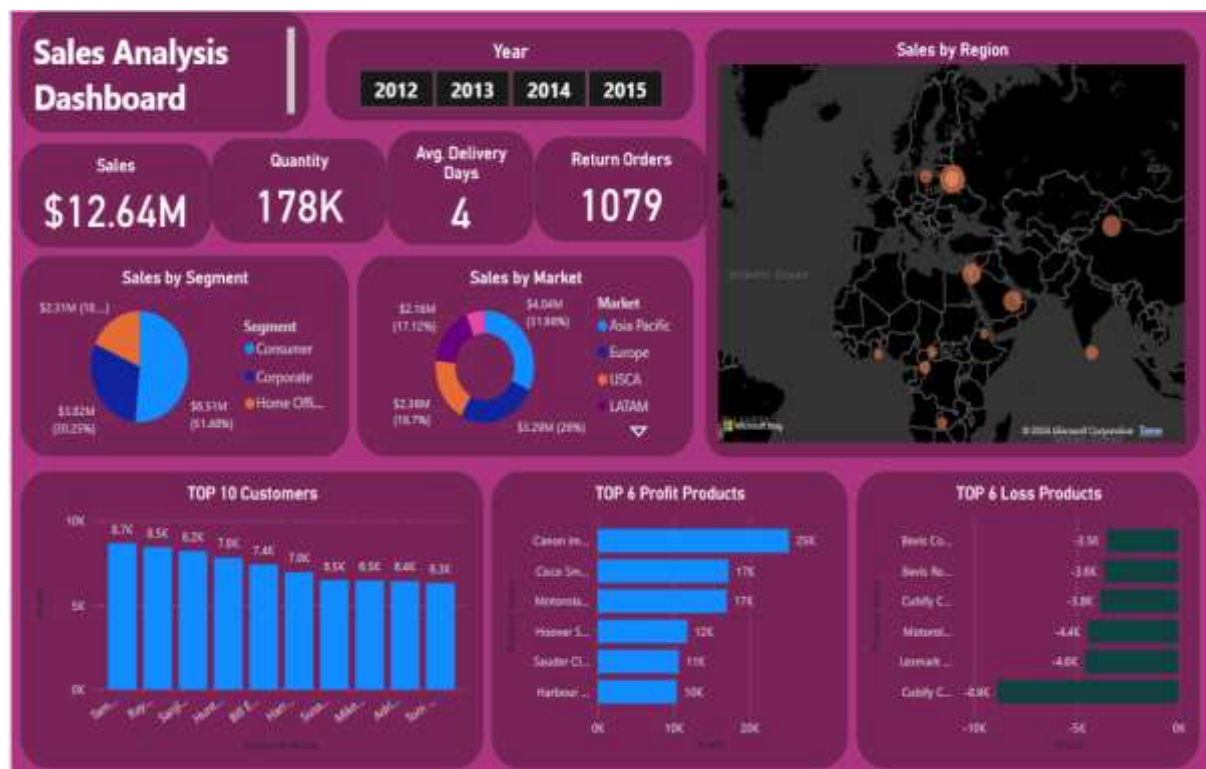
Methodology / Steps Followed

1. Downloaded the **business dataset** from Kaggle for exploratory data analysis.
2. Imported the dataset into the Python environment using the **Pandas** library.
3. Examined the structure of the dataset using functions such as `head()`, `tail()`, `info()`, and `describe()`.
4. Identified the number of rows, columns, data types, and statistical characteristics of numerical variables.
5. Checked the dataset for **missing values and duplicate records**.
6. Performed data cleaning by handling missing values, removing duplicate records, and correcting data types where necessary.
7. Conducted **univariate analysis** to understand the distribution of individual variables such as Sales, Profit, Quantity, and Discount.
8. Performed **bivariate and multivariate analysis** to identify relationships between different business variables.
9. Created visualizations using **Matplotlib and Seaborn** to identify patterns and trends.
10. Analyzed sales and profit performance across different product categories and sub-categories.
11. Examined regional and geographical performance to identify high-performing and low-performing business areas.
12. Analyzed customer segments to understand purchasing behavior and revenue contribution.

13. Identified products or categories generating high sales but low profits.
14. Used correlation analysis to understand relationships between numerical business variables.
15. Identified outliers and unusual patterns that could affect business performance.
16. Summarized the findings and generated meaningful business insights to identify opportunities for improvement.

Output

Name	Data	Item	Kind	Hidden
Orders	Table	Orders	Sheet	FALSE
Returns	Table	Returns	Sheet	FALSE
People	Table	People	Sheet	FALSE
Orders_item_FilterDatabase	Table	Orders_item_FilterDatabase	DefaultName	TRUE
People_item_FilterDatabase	Table	People_item_FilterDatabase	DefaultName	TRUE



Results

- The business dataset was successfully imported and analyzed using Python.
- Exploratory Data Analysis helped understand the structure, distribution, and characteristics of the dataset.
- Missing values and duplicate records were identified and handled during the data preparation process.
- Sales analysis revealed differences in performance across product categories and regions.
- Profit analysis helped identify categories that contributed positively and negatively to overall business profitability.
- Customer segment analysis provided insights into the purchasing behavior and sales contribution of different customer groups.
- Distribution plots helped identify the concentration and spread of sales values.
- Correlation analysis revealed relationships between important numerical variables such as Sales, Profit, Quantity, and Discount.
- Scatter plots helped examine the relationship between sales and profit.
- Outlier analysis identified unusual transactions that may significantly affect overall business performance.
- Data visualization made it easier to identify hidden trends and patterns within the business dataset.

Business Insights

- Certain product categories contribute significantly more to total sales than others, indicating areas of strong customer demand.
- High sales do not always guarantee high profits, highlighting the importance of analyzing profitability along with revenue.
- Some products may generate large revenue but have lower profit margins due to discounts or operational costs.
- Regional analysis helps identify high-performing markets and regions that require improvement.

- Customer segments contribute differently to overall sales, which can help businesses design targeted marketing strategies.
- Discounts may increase sales volume but can negatively affect profitability if not managed properly.
- A small number of high-value transactions can significantly influence overall sales performance.
- Correlation analysis helps identify factors that have a strong relationship with business performance.
- Seasonal or time-based trends can help organizations plan inventory and marketing activities more effectively.
- Underperforming product categories can be reviewed for pricing, promotion, or inventory optimization.
- High-performing regions and customer segments can be prioritized for future business expansion.
- EDA helps transform raw business data into actionable insights that support strategic decision-making.

Conclusion

This experiment demonstrated the application of **Exploratory Data Analysis (EDA)** techniques on a business dataset using Python. The dataset was examined, cleaned, and analyzed to identify important trends, patterns, relationships, and business opportunities.

Various statistical techniques and data visualization methods were used to analyze sales, profit, customer segments, product categories, regional performance, and relationships between business variables. The analysis highlighted the importance of examining both revenue and profitability to gain a complete understanding of business performance.

The insights obtained through EDA can help organizations identify high-performing areas, detect underperforming products or regions, optimize pricing and discount strategies, improve customer targeting, and support effective data-driven decision-making.

Learning Outcomes

- ✓ Understand the importance of Exploratory Data Analysis in business analytics.
- ✓ Import and analyze business datasets using Python and Pandas.

- ✓ Examine dataset structure using head(), info(), and describe().
- ✓ Identify and handle missing values and duplicate records.
- ✓ Perform descriptive statistical analysis.
- ✓ Conduct univariate analysis to understand individual variables.
- ✓ Perform bivariate analysis to identify relationships between variables.
- ✓ Analyze sales and profit performance across different categories.
- ✓ Compare regional business performance.
- ✓ Analyze customer segments and purchasing patterns.

GitHub Link:

https://github.com/lavanyae4319/Data_Analytics_Assignments